

Preprocessing Steps and Business Questions with Visualization

Data Preprocessing Steps

Step 1: Import Dataset

- Import the UNESCO World Heritage Sites dataset into Tableau.

Step 2: Remove Duplicate Records

- Identify and remove duplicate heritage site records.

Step 3: Handle Missing Values

- Replace or remove null values in Country, Region, Category, Area, and Year fields.

Step 4: Correct Data Types

- Convert Year to Date/Number.
- Convert Area to Numeric.
- Ensure Site Name and Country are Strings.

Step 5: Standardize Data

- Correct inconsistent spellings.
- Standardize country and region names.

Step 6: Filter Invalid Records

- Remove incomplete or incorrect records.
- Validate coordinates and area values.

Step 7: Create Calculated Fields

Examples:

- Site Count
- Average Area
- Heritage Age
- Category Percentage

Step 8: Verify Data Quality

- Check for missing values.

- Verify totals.
 - Validate dashboard results.
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Business Questions with Visualizations

Business Question 1

Which countries have the highest number of UNESCO World Heritage Sites?

Visualization: Horizontal Bar Chart

Business Question 2

What is the distribution of Cultural, Natural, and Mixed Heritage Sites?

Visualization: Pie Chart

Business Question 3

How are UNESCO World Heritage Sites distributed across different regions?

Visualization: Filled Map / Symbol Map

Business Question 4

How many World Heritage Sites were inscribed each year?

Visualization: Line Chart

Business Question 5

Which heritage sites are listed as endangered?

Visualization: Highlight Table or Bar Chart

Business Question 6

What is the total area covered by UNESCO World Heritage Sites by category?

Visualization: Treemap

Business Question 7

Which regions have the highest concentration of heritage sites?

Visualization: Bubble Chart

Business Question 8

How has the number of UNESCO World Heritage Sites grown over time?

Visualization: Area Chart

Business Question 9

What are the top 10 countries with the largest heritage site areas?

Visualization: Vertical Bar Chart

Business Question 10

What is the trend and forecast of UNESCO World Heritage Site inscriptions?

Visualization: Forecast Line Chart